

NSC 4356 – Neuropharmacology

Spring 2026

TR 11:00 am – 12:15 pm, SR 224



Instructor	N. Bradley Keele, Ph.D.
Office	BSB A.209
Email	NB_Keele@Baylor.edu
Student	MW 1:30 – 3:00 PM,
Drop-in Hours	or by appointment

[Canvas Inbox or email](#) is the best way to contact me. In most circumstances, I should be able to respond within 1 or 2 hours. Response times on weekends or holidays may be longer, but usually by the next class day.

Course Description:

The purpose of this course is for students to gain mastery of the mechanisms underlying the effects of endogenous and exogenous substances that alter brain function and behavior.

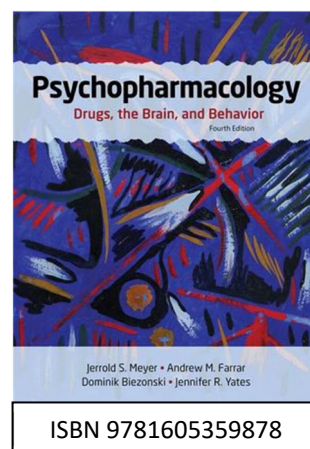
The over-arching goal of this course is for students to gain competency in neuropsychopharmacology by integrating the roles of brain, drugs, and behavior across biological, affective, and cognitive domains. The major course objectives are to:

1. explain the biological basis of drug actions
2. gain detailed understanding of the methodologies used to investigate the pharmacology of the nervous system
3. review/critique empirical evidence of neuropharmacology and drug-brain-behavior relationships
4. critically examine the pharmacology of major neurotransmitter systems
5. analyze and evaluate the pharmacological and neurochemical mechanisms of drugs of abuse, including the affective and cognitive processes supporting chronic substance use
6. evaluate evidence supporting the clinical use of the major psychotropic medications, including mechanisms of drug-mediated changes in affect, cognition and behavior (i.e., clinical efficacy) and common adverse effects

Note. These objectives can be met in several different ways, using a combination of online and in-person instruction. Should events occur that require fewer class days than shown on the course schedule (see last page of the syllabus), we will adjust the format of instruction to meet these goals without adding additional in-person class days.

Required Course Materials:

This course is part of the [Follett ACCESS](#) program where your course materials are billed to your Baylor account as part of your tuition and fees. **Do not purchase a textbook or online access code for this course.** You already have access to your digital materials unless you opted-out of the program through the Baylor Bookstore Customer Portal. If you opted-out by mistake, please [return to the portal](#) and opt-in to regain access to your book. The digital materials are REQUIRED for this course



- Meyer, J. S., Farrar, A. M., Biezonski, D. and Yates, J. R. (2023). *Psychopharmacology: Drugs, the brain, and behavior* (4th ed.). New York: Oxford University Press.

Evaluation of Students:

Final Grade Breakdown

Grades will be determined by performance on the following types of assignments:

Lecture video quizzes	100 points
In-class quizzes (10 highest of 13 quizzes)	200
Unit exams (best 2 out of 3 exams)	300
<u>Comprehensive final exam</u>	300
TOTAL	900 points

Grading Scale: The table below indicates the scale for a specific letter grade for the course.

Course Letter Grade	Points Required
A	806 or more
B	716 – 805
C	581 – 715
F	580 or less

Note: The points required for a specific letter grade MAY change at the end of the semester. There is no guarantee they will change, but if the point requirements change, the change will always be in favor of the student. In other words, the points required for a letter will grade MAY go down, but they will not be increased. Changes to point requirements are at the sole discretion of the instructor.

Lecture Video Quizzes

Prior to each class period, students will prepare for class by watching short lecture videos. The videos are posted to Canvas, and students will log in to Canvas via the device of their choice to access the assignments to view the videos. Embedded in each video are a few questions. Each lecture video assignment must be completed by 11:59 p.m. the night prior to class (i.e., there are assignments due essentially every Monday and every Wednesday night). Late assignments submitted within 24 hours of the due date will receive half credit. Assignments later than 24 hours will not be accepted for credit.



In-class Quizzes



Quizzes will assess your mastery of content from the lessons (both assigned reading and lecture) of that week. Quizzes will be conducted in class during the scheduled class time. The quiz questions are intended to give you frequent assessment and feedback regarding mastery of course content. If used properly, the quizzes may also be a study guide for the semester and final exams.

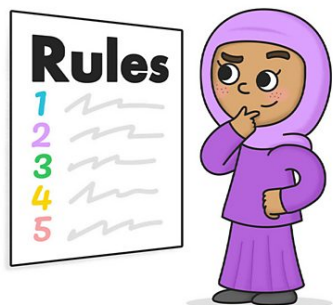
- The highest 10 quiz grades will be used to compute your quiz average.
- Quizzes are typically administered at the beginning of class. **Students MUST be seated in class when the quiz begins to be considered present for the quiz.** If you miss a quiz for any reason (excused, unexcused or otherwise), or if you are late to class and the quiz has already commenced, that quiz grade will be zero because you will drop approximately 3 low quiz grades, including grades of zero.
- Quizzes may include several question formats, such as multiple choice (including true-false), matching, short answer or identify questions.

Unit Exams

The exams given during the semester will be composed of 75 multiple choice questions (MCQs) worth 2 points each. Exam questions will be based on material from assigned reading, lectures, and classroom discussions. Most students find that they must attend class to be party to discussions which prepare them for exam questions. This is especially important for topics not covered in the text. That is, merely reviewing the lecture slides and perhaps someone else's notes are not adequate preparation for the exams. The examinations are not aimed at testing your recognition of important material but are designed to test comprehension and content mastery. Therefore, you should plan on intensive study of the material in this class. Many students spend three or more hours outside class for every hour in class.

Exam Schedule

Exam	Date
Unit 1 Exam Ch 1 – 4, 7 & 8	Tuesday, Feb. 17
Unit 2 Exam Ch 5, 6, 9 & 12	Thursday, Mar. 26
Unit 3 Exam Ch 10, 11, 14, & 17 - 19	Tuesday, May 5
ALL makeup exams See “Makeup Exam Policies” below	Tuesday, May 5 @ 4:30 PM, Room TBA
Comprehensive Final Exam	Saturday May 9, 2:00 – 4:00 PM



Exam Day Rules.

1. The exam closes 10 minutes after class is scheduled to begin. A student arriving after the exam is closed is not permitted to take the exam.
2. You **MUST** submit your exam if you leave the room once you have started the exam. Re-entering the room is not permitted on exam days. Once you leave, the exam is complete. Plan accordingly.
3. Cell phones or other communication / media devices are **prohibited** during exams.

Comprehensive Final Exam

The comprehensive final exam is administered according to the University [final exam schedule](#). The final exam is comprised of 100 questions (three points each), worth 300 points toward your final course grade. The final exam covers material from each section of the course, and there are approximately 33-34 questions from each of the four units.

According to the university [Final Exam Policy](#), “No final examination may be given on other than regularly scheduled dates unless approved in advance by the appropriate academic dean. “A student with 3 or more exams scheduled on a single day may [petition](#) the Dean of the College of Arts & Sciences to reschedule a final exam. If any student chooses to reschedule the Final Exam for this course, the student will receive a grade of incomplete for the current semester, and the final exam will be rescheduled for the first week of classes the following semester.

Makeup Exam Policies:

- If a student misses an exam, a makeup exam may be approved if the student follows the student policy for class attendance (see [Student Policies and Procedures](#), Class Attendance). A makeup exam may be approved if a student misses the exam for University-sponsored activities, participation in (and travel to) an official athletic competition, serious illness, accident, or death in the family.
- Students should seek to arrange with the instructor for a makeup exam prior to the absence when possible.
- Documentation is required to be approved for a makeup exam. Examples include notification from your sponsor or advisor (or other University official), academic coach (SACE), the Chaplain's office ([Spiritual Life](#)), or a physician. The student must provide documentation on the first day you return to class. Without such documentation immediately upon your return to class, no makeup exam will be approved.
- If you miss any of the exams for any reason not outlined in Student Policies and Procedures, you will receive a zero for the missed exam and there will be no makeup exam opportunity. In this case, the missed exam score of zero would be dropped in the calculation of your grade.
- A student may make up only one missed exam. If more than one unit exam is missed, the second missed exam will receive a score of zero.
- **ALL makeup exams will be given on the last class day at 4:30 PM.** The specific location of makeup exams is to be announced later in the semester.
- Students missing the final exam for reasons outlined in Student Policies and Procedures MAY receive a grade of incomplete for the current semester and the final exam MAY be rescheduled for the first week of classes the following semester.
- Extended deadlines to complete assignments will not be granted except in extraordinary, occasional, isolated circumstances. The decision to extend a deadline is completely at the discretion of the instructor. In the event an extension is granted, the time and date of the new deadline is completely at the discretion of the instructor.

COURSE POLICIES

Academic Integrity

Plagiarism or any form of cheating involves a breach of student-teacher trust. This means that any work submitted under your name is expected to be your own, neither composed by anyone else as a whole or in part, nor handed over to another person for complete or partial revision. Be sure to document all ideas that are not your own. Instances of plagiarism or any other act of academic dishonesty will be reported to the Honor Council and will result in failure of the course. Not understanding plagiarism is not an excuse. As a Baylor student, I expect you to be intimately familiar with the [Honor Code](#).

Academic integrity do's and don'ts:

- DO review the Exam Rules
- Lack of understanding acts of academic dishonesty is not an excuse. I expect all Baylor students to be intimately familiar with all aspects of the [Honor Code](#).
- You may use online resources to study for this course, but you must do so in ways that are consistent with all aspects of the Baylor University Honor Code (see, specifically, Section III.C.12 and Section III.C.16).
- You may NOT share quiz or exam information with other students.
- Any and all acts of academic dishonesty MUST be reported to the instructor.
- If you are unsure whether a specific behavior is permissible, ASK your instructor.
Better yet, DON'T DO IT.

Any instance of cheating, plagiarism, or other forms of dishonesty will result in a grade of F for this course.

Baylor University, like any university, is a community of scholars dedicated to pursuing knowledge, using knowledge in service to the community, and communicating knowledge with each other. At this University we all have responsibilities. A professor's responsibility is to teach an academically respectable course using the best available information. Professors are expected to come to class prepared. They must be honest, fair, and objective in their evaluation of students. They must not abuse their power by harassing students or creating unfair advantages. Students also have responsibilities. As a student, your responsibility is to put forth a serious effort to learn this material, to prepare for each class, to engage actively in the lecture material, to remain in class for the duration of the class period, and to participate honestly in all evaluations. Failure on either of our parts is a breach of our implicit academic contract. Breaching this contract should have serious consequences for an individual, be they professor or student, for this basic covenant is the core of university endeavor.

Accommodations

Any student who needs academic accommodations related to a documented disability should inform me immediately at the beginning of the semester. You are required to obtain appropriate documentation and information regarding your accommodations from the Office of Access and Learning Accommodation (OALA). OALA is located in the basement level of the east wing of the Sid Richardson building (reachable via elevator or stairs). They can also be reached by phone (254) 710-3605 or email OALA@baylor.edu.

All accommodations should be handled through the [Office of Access and Learning Accommodation](#). Students who desire to use accommodations for this course must send a letter notifying the instructor of the eligible and approved accommodations to be used. To have access to your accommodation(s) and for OALA to send your letter, you must make this request at the start of each academic term. Accommodations are not retroactive and only apply forward once your letter has been sent to the instructor. Requests for notification letters and for testing accommodation (distraction-free testing, alternate test location, extra time) must be completed using the [Accommodate](#) platform. Students must comply with all deadlines and guidelines as determined by the OALA.

Attendance

Successful students maintain regular attendance in this course, arriving promptly to class and ready with the necessary materials for the day's work. My attendance policy is detailed below, and all students enrolled will be held responsible for these expectations.

The University [Class Attendance](#) policy and the [Attendance Policy](#) of the College of Arts & Sciences are in effect for this course.

To earn course credit in this course, a student must attend at least 75% of all class meetings. Any student who does not meet the $\geq 75\%$ attendance minimum will receive a grade of “F” in the course. Any University-related activity necessitating an absence from class shall count as an absence when determining whether a student has attended the required 75% of class meetings.

Therefore, if you miss **8 or more classes** you will be assigned a grade of F for this course. To be counted present, you must be in class when attendance is taken. Also, since enforcing the attendance policy includes forwarding mid-semester deficiency reports to the registrar, you will be receive a deficiency report if you have missed three or more classes by the 6th week of class.

Cell Phones and Other Electronic Devices

Please ensure your cell phone or other device (e.g., smartwatch, laptop, tablet, etc.) is not disruptive during class. This may mean placing it in silent (not vibrate) mode or turning it off. I am modestly comfortable with the appropriate and responsible use of electronic devices during class. There may be times at which use of your phone and/or computer can promote academic goals. We will use media devices to access quizzes in class. There are also numerous online resources, and it is pedagogically valuable to use those resources for additional information about the current discussion topic. However, if you become a distraction to those around you (texting, social media, games, etc.), there are consequences. On the first infraction, you will be given a warning (and only one warning). Upon further misuse of electronic devices, I will ask you to leave the classroom. **Cell phones are absolutely PROHBITED during exams.**

Class Format Conversion

This course may need to be converted to a hybrid or online format at some point in the semester. Your instructor has anticipated this possibility and prepared contingency plans for the course. Necessary information will be provided to you in the event we no longer meet in-person.

Classroom Atmosphere

We want to create a classroom atmosphere in which we are able to express ourselves in an open and honest way while being sensitive to others. It's ok to get emotional, to disagree, and to debate. However, let's do our best to avoid both macro- and micro-aggressions¹. We must recognize that our intent doesn't always match our impact and that despite our best efforts, our actions and words may at times be experienced negatively. The commitment to lifelong learning in the profession-wide competencies includes constantly working to improve these skills. Thus, if aggressions occur, let's do our best to learn and grow from these experiences by bringing them to our attention and suggesting alternatives. While the goal is to be able to gently assist each other “in the moment,” this can be difficult. Thus, if you are uncomfortable, I invite you to contact your professor or any trusted faculty member. There is also the Baylor University [Report It](#) website, which directs faculty, staff, students, alumni, parents and the public to a range of reporting tools and instructions. If you are unable to identify the appropriate mechanism for your report, please use [EthicsPoint](#). EthicsPoint is an anonymous, confidential hotline/website to report issues for investigation.

¹Microaggression is discrimination that is covert and subtle, whether intentional or unintentional; macroaggressions are overt forms of discriminatory aggression that are obvious (Pierce, 1970).

Classroom Etiquette

Class begins promptly. This means you should be in your seat ready for class at the scheduled starting time. Important class announcements and reminders occur at the beginning or end of class. Because our time together is short each day, respect your peers and your instructor by coming to class on time and refrain from leaving class early. Please do not pack up bags or stand before you are dismissed. The instructor will wrap up at the designated ending time and dismiss the class.

Please turn off or silence notifications on all phones, smartwatches, smart glasses, and similar devices before class begins. **Phones and all other electronic devices are PROHIBITED during exams.**

BAYLOR STUDENT RESOURCES

Academic Success

I believe every student who has been admitted to Baylor can be successful, and I want to partner with you to help you thrive academically. Be sure to take advantage of the many resources available for academic success, including coming to see me during my office hours. Students who regularly utilize the great resources in the [Paul L. Foster Success Center](#) such as [tutoring](#), [Learning Lab](#), and [Academic Mentoring](#) are among my most successful students. If your academic performance in this class is substandard, I will submit an Academic Progress Report to the Success Center so that the team of coordinated care professionals can ensure that you get the help you need.

Crisis and Emergency Numbers

Please make a note of the following numbers for crises or emergencies:

- **Counseling Center Crisis Line:** (254) 710-2467 (Business Hours/Non-Business Hours/Weekends)
- **Baylor Police Department:** (254) 710-2222
- **MHMR Crisis Center:** (254) 867-6550
- **MHMR 24-Hour Emergency/Crisis Number:** (254) 752-3451
- When home during academic breaks, when the counseling center is closed, please call your local resources and/or national hotlines:
 - National Hope Network Hotline: 1-800-SUICIDE (1-800-784-2433)
 - National Suicide Prevention Lifeline: 1-800-273-TALK (1-800-273-8255)

First-Generation Students

Baylor University defines a first-generation college student as a student whose parents did not complete a four-year college degree. The First in Line program is a support office for first-generation college students to utilize if they have any questions or concerns. Please check out [First in Line](#), visit us in Sid Richardson, or email firstinline@baylor.edu.

Military-Connected Students

Veterans, active-duty military personnel, and dependents are encouraged to connect with the [VETS](#) program, a space dedicated to supporting our military-connected students. Please communicate, in advance, if possible, any special circumstances (e.g., upcoming deployment, drill requirements, disability accommodations).

Sexual and Interpersonal Misconduct Policy and Civil Rights Policy

Baylor University does not tolerate unlawful harassment or discrimination on the basis of sex, gender, race, color, disability, national origin, ancestry, age, citizenship, genetic information or the refusal to submit to a genetic test, past, current, or prospective service in the uniformed services, or any other characteristic protected under applicable federal, Texas, or local law (collectively referred to as Protected Characteristics). These policies also prohibit discrimination and harassment based on pregnancy or related conditions.

If you or someone you know would like help related to an experience involving:

1. Sexual assault, sex-based harassment or discrimination, dating violence, domestic violence, stalking, sexual exploitation, or retaliation for reporting one of these types of prohibited conduct, please visit www.baylor.edu/titleix, or contact us at (254) 710-8454, or TitleIX_Coordinator@baylor.edu.
2. Harassment or discrimination (excluding those issues listed in #1) based on Protected Characteristics, please visit www.baylor.edu/civilrights, or contact us at (254) 710-8454 or Civil_Rights@baylor.edu.

For pregnancy or related conditions, the University will provide reasonable modifications to policies, practices, or procedures as necessary to ensure equal access to educational programs and activities. Individuals who need to request reasonable modifications should contact the [Office of Access and Learning Accommodation](#). More information on pregnancy and parenting resources can be found [here](#).

The Equity, Civil Rights, and Title IX Office understands the sensitive nature of these situations and can provide information about available on- and off-campus resources, such as counseling and psychological services, medical treatment, academic support, university housing, and other forms of assistance that may be available. Staff members at the office can also explain your rights and procedural options. You will not be required to share your experience. **If you or someone you know feels unsafe or may be in imminent danger, please call the Baylor Police Department (254-710-2222) or Waco Police Department (9-1-1) immediately.**

Except for Confidential Resources, all University Employees are designated Responsible Employees and thereby mandatory reporters of potential sexual and interpersonal misconduct violations. Confidential Resources who do not have to report include those working in the Counseling Center, Health Center, Athletics Mental Health, and Baylor Psychology Clinic, and the University Chaplain and LHSON Chaplain.

Student Wellbeing

I recognize that students face many struggles during their Baylor career. [The Department of CARE Team Services](#) is a group of dedicated and caring case managers who work with students facing mental health issues, financial struggles, and anything else affecting a student's ability to be successful in the classroom. You can find them on the second floor of the Student Life Center, suite 207. Contact Information: (254) 710-2100 ; CareTeam@baylor.edu.

Health Care

Baylor University is strongly committed to addressing the physical well-being and mental health of students by providing access to on-campus healthcare resources. [Baylor Health Services](#) includes Primary Care, Psychiatry, Physical Therapy and Pharmacy and is staffed with fully certified and licensed physicians and nurse practitioners, as well as nurses and administrative staff. Appointments may be made by calling their main number or by logging into the health portal located on their website. Contact Information: (254) 710-1010; Health_Services@baylor.edu.

Mental Health

You also have access to mental health services through [The Counseling Center](#). Connect with them for an initial appointment, goal-directed individual therapy, group therapy, nutritional counseling, and case management services. You can schedule your 30-minute initial appointment online through the [health portal](#) or by calling (254) 710-2467. If you are experiencing suicidal thoughts, call or text #988 anytime day or night for support.

Telehealth

All Baylor students also have access to medical and counseling services via telehealth in addition to the on-campus services. Medical services include virtual urgent care, psychiatry, and nutrition counseling. Counseling services include 24/7/365 in-the-moment support and ongoing mental health counseling. Access to care is available in the evenings, during weekends, and when the University is closed. All services are free and unlimited to Baylor students. Please visit <https://baylor.academiclivecare.com/> for more information.

Addiction

[The Beauchamp Addiction Recovery Center](#) (BARC) supports students in recovery from substance and behavioral addictions through an all-encompassing level of support approach that includes one-on-one mentorship, support groups, and social events open to all Baylor students. Located in the East Village Residential Community (bottom floor of Teal Residential College). Contact Information: (254) 710-7092; BARC@baylor.edu

Food Insecurity

At Baylor, we want all students to have access to food resources that will support their holistic well-being and success. If you or someone you know experiences food insecurity at any time, you can find information on campus and community food resources by visiting [The Store](#). You can also contact Store staff at (254) 710-4931. For additional basic needs assistance, please reach out to [CASE](#) or the [Care Team](#).

Spirituality

[The Office of Spiritual Life](#) offers programs, persons, and resources to nurture theological depth, spiritual wholeness, and missional living. Located on the corner of 5th and Speight Street in the Bobo Spiritual Life Center. Contact Information: (254) 710-3517; Spiritual_Life@baylor.edu.